# Zoom In, Scale Out

*Metric depth model gulo ki digital zoom er physics mane?*

*Research overview · Rabeya Noor · target CVPR 2027 · 19 October 2026*

**Ek line e:** Chhobi crop kore boro korle (digital zoom) focal length thik s gun bare, kintu kono jinish kachhe ashe na. Amra dekhiyechi beshirbhag metric depth model zoom korle depth kom bole: shobcheye popular Depth Anything V2 3x zoom e depth prai 1/3 bole (α = 0.95). Ete label lage na, theory ache, ar label chhara fine-tuning (EFT+B) diye eta prai puro thik kora jay.

## 1. Shomossha

Self-driving, robot, AR ar phone camera te ekta chhobi theke protita jinish koto **meter** dure, sheta bolte metric depth model lage (Depth Anything, Depth Pro, MoGe, UniDepth, Metric3D, Depth Anything 3). Meter e bolte hole camera er focal length jana dorkar. Karon wide lens e kachher jinish ar telephoto lens e durer jinish ekdom ek rokom dekhate pare. Model gulo camera chhara shudhu pixel dekhe eta anuman kore. Proshno holo: ora ki asholei camera bujhe, naki shudhu "boro dekhale kachhe" shortcut use kore?

## 2. Mul idea: digital zoom ekta free, shothik ruler

Chhobir ekta box crop kore abar ager size e resize korle (digital zoom), chhobi ta ekdom shothik bhabe ekta notun pinhole camera theke tola chhobir moto hoy: f' = s·f, principal point shore jay, kintu **depth bodlay na**. Tai label chhara test kora jay: zoom er age ar pore model nijer depth ar focal er sathe tulona korlei hoy.

| Shonkhya | Mane | Thik model | Shortcut model |
|---|---|---|---|
| α (depth exponent) | D̃(zoom) / D̃(original) ≈ s^−α | 0 | 1 (2x zoom e depth ordhek) |
| β (focal exponent) | f̃(zoom) / f̃(original) ≈ s^β | 1 | 0 (focal bodlay na) |

## 3. Theory

### 3.1 Tinta invariant

- α = 0 → depth thik thake;
- β = 1 → 3D shape thik thake;
- α + β = 1 → shudhu pasher (lateral) map thik thake, depth ba shape na. "Boro dekhale kachhe" reasoning thik ei line e pore: Ḋ = f̃·L/ℓ hole zoom e Ḋ ∝ s^(β−1).

### 3.2 Zoom–squash ambiguity (notun)

Focal s·f e tola chhobi pixel-by-pixel sheyi scene er chhobi, jeta depth er dik borabor s gun chepta kora: u = s·f·X/Z = f·X/(Z/s). Chepta scene e jinish er width ek thake, shudhu gobhirota bhul hoy (4.5 m er gari 2x e 2.25 m lomba hoye jay). Tai shudhu size cue diye zoom dhora oshombhob, 3D shape prior lage. Ar shoja dewal (fronto-parallel plane) e kono model-i camera na jene zoom dhorte parbe na.

### 3.3 Stale K

Metric3D er moto camera-input model e zoom er por purono K dile network ekoi pixel dekhe, shudhu shesh e scale bodlay. Tai α(stale) = α(true) + 1, eta ganit diye proman kora jay. Data te median tofat 1.000.

## 4. Experiment setup

| Ki | Details |
|---|---|
| Model (main) | DA-V2-S/B, ZoeDepth (pixels only) · Depth Pro, MoGe-2-S, UniDepthV2-S (predicts camera) · Metric3D v2-S, thik K ar stale K (takes camera) |
| Model ladder | DA-V2-L, MoGe-2-B/L, UniDepthV2-B/L, UniK3D-L, Metric3D v2-L, Depth Anything 3 (Nested-Giant, Metric-L, L), VGGT-1B |
| Data | KITTI val selection (LiDAR), NYUv2 test (Kinect), DIODE val indoor + outdoor (laser). Protita theke 200 chhobi, shob model e ekoi chhobi (800 chhobi/model) |
| Zoom | 1, 1.25, 1.5, 2, 3x majhkhane · off-centre: 4 kona, 50 chhobi (KITTI, NYU) |

| Ki | Details |
|---|---|
| Control | blur control (zoom er moto jhapsa, field of view ek) · geometry check (pixel error 0) · unzoomed prediction ekoi pixel e (fair reference) · result verify tool (56 file, shob check pass) |
| Statistics | 95% bootstrap CI · paired Wilcoxon signed-rank + Holm correction · Spearman with bootstrap CI |
| Compute | Kaggle T4 GPU; laptop e shudhu analysis |

## 5. Result 1: zoom audit

| Model | Group | α [95% CI] | β | α+β | Blur control | AbsRel 1x | AbsRel 3x |
|---|---|---|---|---|---|---|---|
| DA-V2-S | pixels only | 0.95 [0.93, 0.99] | – | – | 1.3% | 0.317 | 0.652 |
| DA-V2-B | pixels only | 0.85 [0.82, 0.87] | – | – | 0.8% | 0.313 | 0.599 |
| ZoeDepth | pixels only | 0.61 [0.58, 0.64] | – | – | 0.7% | 0.310 | 0.539 |
| Depth Pro | predicts camera | 0.31 [0.27, 0.34] | 0.65 | 0.97 | 2.8% | 0.310 | 0.459 |
| MoGe-2-S | predicts camera | 0.36 [0.34, 0.38] | 0.56 | 0.94 | 3.1% | 0.274 | 0.465 |
| UniDepthV2-S | predicts camera | 0.34 [0.30, 0.40] | 0.45 | 0.87 | 4.0% | 0.824 | 0.792 |
| Metric3D v2-S | takes camera (true K) | -0.07 [-0.08, -0.06] | – | – | 1.6% | 0.142 | 0.358 |
| Metric3D v2-S (stale K) | takes camera (stale K) | 0.93 [0.92, 0.94] | – | – | 1.6% | 0.142 | 0.628 |

**Ja pelam:**

- Pixels-only model gulo prai puro shortcut use kore: DA-V2-S α = 0.95.
- Camera predict kora model gulo ordhek bujhe: α ≈ 0.31–0.36, ar focal 3x er jaygay 1.63–2.04x bole.
- Depth Pro e protita chhobite α+β er median 0.97, mane theory er "lateral invariant" line e prai hubohu thake.
- Thik K pele Metric3D prai equivariant (α = -0.07), stale K dile pixels-only model er moto (0.93).
- Blur control e depth bodlay mattro 0.7–4%, tai dosh interpolation er na, camera bojhar.

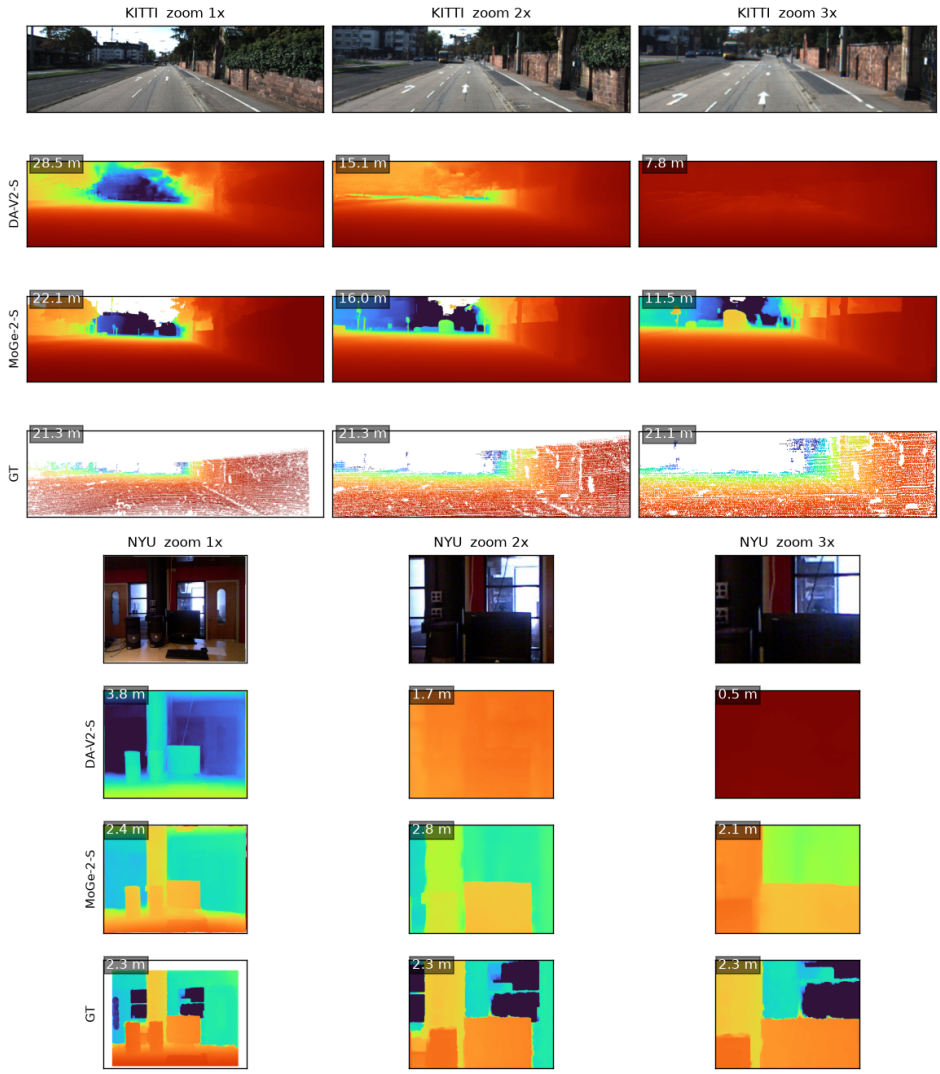



*Figure 1. Ekoi scene 1x, 2x, 3x zoom e. Ek row er shob depth panel ekoi rong er scale e (meter). DA-V2-S zoom korle "kachhe" chole jay (KITTI 28.5 → 7.8 m, GT 21 m); MoGe-2 ordhek.*

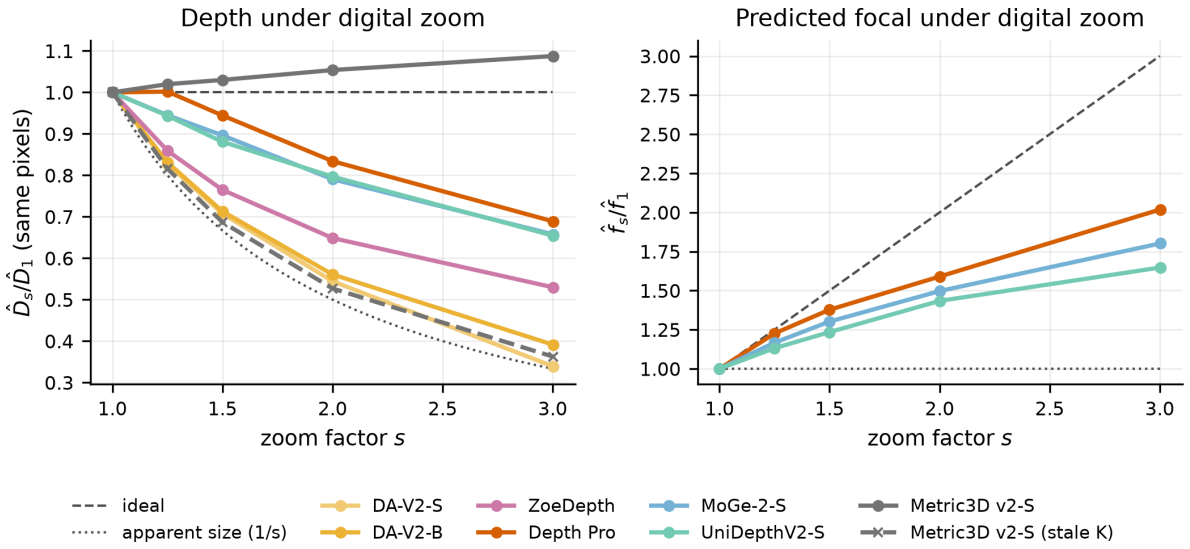



*Figure 2. Zoom e depth (bame) ar predicted focal (dane), 800 chhobir median.*

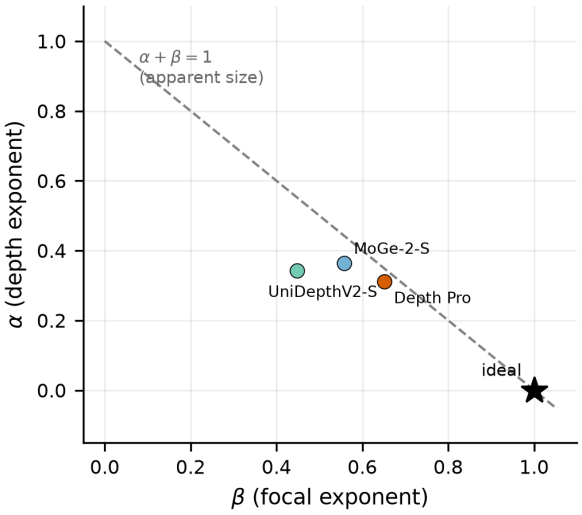



*Figure 3. α–β plane. Dashed line: α+β = 1 (apparent size reasoning). Tara: ideal.*

## 6. Result 2: label chhara score diye error anuman

|α| (label lage na) ar 3x zoom e AbsRel er Spearman ρ = 0.67 [0.32, 0.90], 28 ta model×dataset jora. Dataset er bhitore KITTI 0.96, NYU 0.96, DIODE-in 0.86, kintu DIODE-out -0.14. Shadharon (1x) accuracy predict kore na (ρ = -0.06). **Shotti kotha:** zoom e depth shrink korle error bare-i, tai sambandha kichuta nijer theke ashe. Claim ta tai choto: "zoom e kon model koto bhul korbe, label chhara bola jay."

## 7. Result 3: label chhara fine-tuning (EFT ar EFT+B)

Frozen copy (teacher) zoom chhara chhobi te ja bole, student ke zoom kora chhobi te ta-i bolte shekhai. Kono depth label ba camera info lage na. Training data: 7 ta KITTI raw drive (test drive er sathe overlap nai, code nijei check kore) ar NYUv2 er 795 ta train chhobi. DIODE kokhono training e dekhano hoy na.

**Notun finding, resolution shortcut:** shudhu zoom loss diye train korle (EFT) model shikhe ney "jhapsa chhobi = zoom". Blur control e KITTI te depth 30% bodle jay. EFT+B te ekta blur-invariance term jog kori: jhapsa (kintu zoom na) chhobite ager depth-i bolte hobe. Eta Zhong et al. (ICCV 2023) e nai.

| Dataset | Model | α | AbsRel 1x | AbsRel 3x | Blur shortcut |
|---|---|---|---|---|---|
| KITTI | DA-V2-S (frozen) | 1.03 | 0.138 | 0.665 | 1.1% |
| KITTI | EFT | 0.18 | 0.156 | 0.158 | 29.6% |
| KITTI | EFT+B (ours) | 0.05 | 0.175 | 0.178 | 4.9% |
| NYUv2 | DA-V2-S (frozen) | 0.98 | 0.344 | 0.563 | 0.7% |
| NYUv2 | EFT | 0.12 | 0.313 | 0.226 | 4.2% |
| NYUv2 | EFT+B (ours) | 0.07 | 0.301 | 0.266 | 0.4% |
| DIODE-in (unseen) | DA-V2-S (frozen) | 0.70 | 0.280 | 0.558 | 1.0% |
| DIODE-in (unseen) | EFT | 0.39 | 0.368 | 0.364 | 0.6% |
| DIODE-in (unseen) | EFT+B (ours) | 0.37 | 0.400 | 0.367 | 0.4% |
| DIODE-out (unseen) | DA-V2-S (frozen) | 0.65 | 0.731 | 0.551 | 0.6% |
| DIODE-out (unseen) | EFT | 0.39 | 0.877 | 0.648 | 1.5% |
| DIODE-out (unseen) | EFT+B (ours) | 0.40 | 1.000 | 0.676 | 0.4% |

- EFT+B prai puro equivariant (KITTI ar NYU te α 0.05–0.07), ar blur shortcut prai chole geche.
- Trade-off: shadharon (1x) chhobite accuracy kichuta kome, beshi kome DIODE e (training e dekhe nai). λ = 3 (shokto anchor) diye eta kombe kina, sheta ekhon Kaggle e test hocche.
- "Anchor-only" baseline asole frozen model (student = teacher, tai gradient 0); er bodole blur-only baseline cholche.

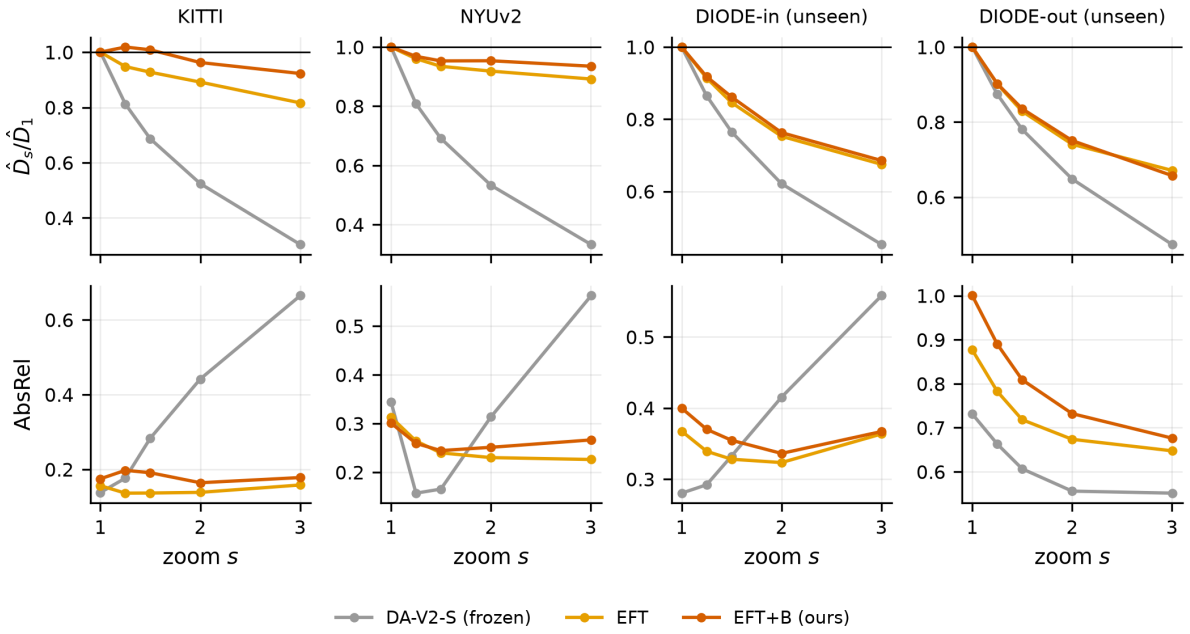



*Figure 5. Upore: zoom e depth ratio (1 = ideal). Niche: AbsRel. Dhushor: frozen, holud: EFT, lal: EFT+B. Note: alada zoom e AbsRel alada pixel e mapa; NYU te frozen model 1.25–1.5x e 1x er cheye bhalo AbsRel pay, karon 1x e depth beshi bole ar zoom shrink kakatale thik er kachhe ane. Tai AbsRel na, α dekhte hobe.*

## 8. Prathomik: model boro hole ki thik hoy? (1 ta chhobi, final na)

| Model | 3x e depth ratio (thik = 1) | 3x e focal ratio (thik = 3) |
|---|---|---|
| DA3-L | – (relative depth) | 1.55 |
| DA3-Metric-L (thik K deya) | 1.07 | – |
| DA3-Nested | 0.38 | 1.18 |
| DA-V2-L | 0.45 | – |
| MoGe-2-B | 0.66 | 2.07 |
| MoGe-2-L | 0.71 | 2.17 |
| UniDepthV2-B | 0.55 | 1.61 |
| UniDepthV2-L | 0.58 | 1.73 |
| UniK3D-L | 0.59 | 2.07 |
| VGGT-1B | – (relative depth) | 2.50 |

Shobcheye notun ar boro DA3-Nested (≈1.4B) 3x zoom e depth 0.38 bole ar focal prai bodlay na. Mane boro ar notun model holei zoom bujhe na. Eta shudhu 1 ta KITTI chhobi; puro run (100 chhobi × 4 dataset) Kaggle e cholche.

## 9. Shotti negative result

- Theory er "soft" prediction (scene e depth beshi chhoriye thakle α kombe) data te pawa jay nai (camera model e ρ ≈ 0).
- Label-free score shadharon accuracy predict kore na, shudhu zoom e accuracy.
- UniDepthV2-S DIODE-outdoor e lomba error tail dey (prai 10% chhobite 6x beshi depth); α er upor eta prabhab felbe na.
- Stale-K result notun na (Kim & Lee 2026 er kaj ke support kore).

## 10. Agerkar kaj ar amader notun ongsho

| Paper | Ki koreche | Amader sathe tofat |
|---|---|---|
| Zhong et al., ICCV 2023 | crop-resize equivariance, self-consistency loss | camera/metric nai; amra α, β, theory, notun model, shortcut + EFT+B |
| Kim & Lee, Electronics 2026 | stale intrinsics e camera-input model er bhul | amader stale-K tader support kore |
| Roll-robust depth (2026) | camera roll | roll e intrinsics bodlay na, zoom e bodlay |

| Paper | Ki koreche | Amader sathe tofat |
|---|---|---|
| KPE (2023), PlaneDepth (CVPR 2023) | crop er perspective ambiguity | test/training input; zoom audit na |
| FS-Depth (2023) | crop diye focal augment, label shoho | amader label ar focal input chhara |
| DepthCues (CVPR 2025) | depth cue probing (size shoho) | amra size cue er physical meaning (α) |

**Amader notun ongsho:** (1) metric depth er zoom audit with α ar β; (2) α+β = 1 line ar zoom–squash theory; (3) teen dhoroner model (pixels only / predicts camera / takes camera) ek scale e; (4) label-free proxy; (5) resolution shortcut finding ar EFT+B; (6) principal point test ar ZoomFix (EXIF zoom factor diye training chhara fix); (7) model ladder (DA3, VGGT, UniK3D shoho).

## 11. Code

- GitHub (private): github.com/rabeyanoor/zoom-equivariance
- 25 ta unit test (zoom geometry, audit toy model, analysis, ZoomFix), ruff lint, GitHub Actions CI
- Vectorised analysis (loop er cheye 150 gun druto, test e hubohu ek)
- verify_results.py: protita CSV te shob zoom, duplicate nai, shob model ekoi chhobi
- Paper er shob number make_paper_numbers.py diye CSV theke, haate kichu lekha hoy na
- Protita Kaggle run er log logs/kaggle/ e

## 12. Baki kaj

| Kaj | Obostha |
|---|---|
| audit-a2: final 7 model (same-pixel reference, principal point, ZoomFix column) | Kaggle e cholche |
| Model ladder c1 (DA-V2-L, MoGe-2-B/L, UniDepthV2-B/L, UniK3D-L) | Kaggle e cholche |
| Model ladder c2 (DA3 ×3, Metric3D-L, VGGT) | GPU slot er opekkha |
| EFT ablation: blur-only, EFT+B λ = 3 | GPU slot er opekkha |
| Abstract, intro, results, conclusion asol number diye | result ashle |
| (Optional) phone e 1x/2x/5x zoom chhobi, tape diye mapa dure | tumi korbe |
| Overleaf e PDF, supervisor review, CVPR submit (~7–16 Nov) | shesh e |